

PYROGEN TESTING IN PARENTERALS BY LIMULUS AMEBOCYTE LYSATE (LAL) ASSAY

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ABSTRACT

Microslide method of Limulus Amebocyte Lysate (LAL) was used to detect pyrogenic substances in twenty five large volume parenterals. Only one sample exhibited positive LAL test. It is the most economical method because a single drop of lysate is required to detect the presence of pyrogenic substance.

INTRODUCTION

The first true understanding of the so called injection fever was reported by Hort and Penfold in 1912 (Hort and Penfold, 1912). These workers were the first to design and standardize a rabbit pyrogen test. It gained little attention until the work of Seibert which was reported in 1923 (Seibert 1923, 1925). She demonstrated convincingly that injection fever was the result of contamination with pyrogens, filterable heat stable substances that were produced by Gram negative bacilli. To ascertain the presence or absence of febrile responses caused by her test substances, Siebert selected rabbit as her animal test model, a choice that was later proven to be fortuitous (Seibert and Mendel, 1923).

For 37 years the rabbit pyrogen test was the only method used for detecting pyrogenic substances. Then Levin and Bang (Levin and Bang, 1964) introduced in vitro Limulus Amebocyte Lysate test (LAL).

The application of the Limulus Amebocyte Lysate (LAL) test to parenterals arose because a suitable means of detecting pyrogens in short-lived radioactive drugs was needed. Following a decade of use as a pyrogen screen, the LAL test is now established as a quality assurance measure for parenterals because of its simplicity, sensitivity, specificity and reproducibility. Today, pharmaceutical industry is utilizing the new approaches in different fields primarily for improving drug quality through in-

process testing and for trouble shooting specific problems.

A guide line was released by the U.S. Food and Drug Administration in January 1980 [Fed. Reg., 45(13)] to inform manufacturers of human drugs and biologicals, animal drugs and medical devices, about the procedure involved in LAL testing. The agency considers it necessary to validate the use of LAL as an "end-product endotoxin test". Those who adhere to the LAL guideline are considered in compliance with relevant CGMP provisions for drugs and devices. The general endotoxin limit for parenteral drugs is 0.5 EU/Kg dose except for a 0.2 EU/Kg limit for intrathecal drugs. As a result of comments received, several major changes were incorporated in a second draft guideline, issued in February 1983 (Anonymous 1983).

Further development in the LAL technique have resulted in its combination with enzyme linked immunosorbent assay (ELISA). This has provided a useful tool for quantitation of the amount of endotoxin in a sample (Zhang et al., 1988).

The present study has been designed to evaluate the presence of pyrogenic substances in large volume parenterals available in the local pharmacies, using Limulus Amebocyte Lysate (LAL). It is hoped that the present work will provide a firm basis in introducing and establishing LAL test to national and multinational pharmaceutical industries for the detection of pyrogenic substances in parenterals.

EXPERIMENTAL

Twenty five samples of commonly available, locally manufactured and imported LVPs of various manufacturers were procured from different pharmacies of Karachi. The date and place of samples purchased and batch numbers of each were recorded (Table 1). The lyophilized kit of LAL reagent, "PYROGEL"[®] was procured from M/s. CONCEPT GmbH, Germany.

Preparation of Samples:

All operations in collection and preparation of samples were performed aseptically using depyrogenated apparatus and non-pyrogenic reagents as recommended in British Pharmacopoeia 1988 (Addendum 1989). Samples were diluted according to U.S. Food and Drug Administration (FDA) screening procedures to make it compatible with the test. The Maximum valid dilution (MVC) calculated has been reported in Table 2.

Determination of the labeled sensitivity of the lysate:

The sensitivity of the lysate as calculated by standardized method was equal to the labeled sensitivity. It was expressed in EU/ml.

Methodology:

Among various reported methods of endotoxin (pyrogen) detection with LAL assay, following was used during the present study:

(1) Microslide method:

This method was first used by Frauch in 1974. In this method 10 μ l of lysate was placed on the surface of the pyrogen-free glass slides and equal amount of sample was added. The slides were incubated at 37°C for 30 minutes in a water bath. Gel formation was observed by gently moving the slides against the black background.

RESULTS AND DISCUSSION

In the present study the screening of twenty five (Table 2) infusion samples (including dextrose 5%, sodium chloride 0.9%, mannitol 20%, Hartmann (compound sodium lactate) solutions, dextrose saline solutions, metronidazole injections, amino acids solutions, haemodialysis solution, sterile water for injection manufactured by different companies and distilled water from the storage tank of laboratory of Department of Pharmaceutics) has been carried out for the detection of pyrogen using Limulus Amebocyte Lysate test.

Before proceeding for tests, sensitivity of lysate used was determined and tabulated as Table 1.

Table 2 indicates the results of microslide method.

The microslide procedure used during the present study was slide test method (Table 1). Frauch (Frauch, 1974) in 1974 was the first to suggested this simple slide test method using 10 μ l of lysate and equal amount of sample to be tested. Results obtained presently were similar to those achieved by Frauch. The importance of this method lies in the fact that, very minute quantities of lysate and samples are required to evaluate the presence of pyrogenic substances which makes it very economical.

After Frauch, Goto and Nakamura (Goto and Nakamura 1979) developed a slide method using 20 μ l lysate and employing a novel end point. Lysate and samples were mixed on silicone coated glass slides and incubated at 37°C for 30 minutes. A Japanese worker Okuguchi (Okuguchi 1978) reported an improved LAL microslide method employing 10 μ l of lysate. Flowers (Flowers 1979) also reported a modified slide method. Malver and Fystro (Melvaer and Fystro 1982) developed a micro method based on earlier work of Frauch (Frauch 1974) and Flowers (Flowers 1979). However the former workers treated microscope slides with silicone, thereby increasing 4 to 12 times the number of tests that can be done on a single slide.

Table 3 contains results of positive and negative controls, run side by side with the sample. A positive control consists of pyrogen-free water spiked with known concentration of endotoxin and a negative control carried the pyrogen-free water and lysate only. All tests were run in duplicate.

The results also confirmed the simplicity and rapidity of LAL test as it allows more frequent testing, thus providing an added measure of assurance in pyrogen testing programme. Due to these extra advantages the test has been officially accepted in many countries of the world.

It is expected that the present study will be helpful in implementing and establishing the LAL test in different pharmaceutical industries in Pakistan. The technique is being used for detection of endotoxin in clinical samples and its efficacy as an indirect method of detection of gram negative bacteria in clinical samples is being tested (Miller et al., 1990 and McCartney et al., 1989).

Table 1
Determination of labelled LAL sensitivity
Slide method

Replicate No.	Endotoxin concentration EU/ml				End Point
	0.5	0.25	0.125	0.0625	
1.	+	+	-	-	0.25
2.	+	+	+	-	0.125
3.	+	+	-	-	0.25
4.	+	+	-	-	0.25

Geometric mean: 0.25

Lysate sensitivity: 0.25 EU/ml

Table 2
Results of microslide method

S. No.	Drug Product/ Labelled Strength	Manufacturer Name	Batch Number	pH Adjustment	MVD*	Results	
						I	II
1.	Normal saline 0.9%	OTSUKA	214 J91	Not required	4	-	-
2.	Normal saline 0.9%	B. BRAUN	049242	Not required	4	-	-
3.	Normal saline	MEDIPAK	109205	Not required	4	-	-

Table continued

S. No.	Drug Product/ Labelled Strength	Manufacturer Name	Batch Number	pH Adjustment	MVD*	Results	
						I	II
4.	Normal saline	SWEPAK	10526A	Not required	4	-	-
5.	Dextrose 5%	OTSUKA	111 E91	Required	2	-	-
6.	Dextrose 5%	B. BRAUN	M108442	Required	2	-	-
7.	Dextrose 5%	MEDIPAK	110264	Required	2	-	-
8.	Dextrose 5%	SWEPAK	10489A	Required	2	-	-
9.	Mannitol 20%	OTSUKA	20L 86	Required	4	-	-
10.	Osmofundin 20%	B. BRAUN	M030443	Required	4	-	-
11.	Aminovel L600 5%	OTSUKA	21H 94	Required	68	-	-
12.	Aminoplasma L5	B. BRAUN	013262B	Required	68	-	-
13.	Nutrisol 5%	GREEN CROSS	176HS	Required	68	-	-
14.	Perisolution	OTSUKA	256 K91	Required	4	-	-
15.	Peritofundin	B.BRAUN	M908144	Required	4	-	-
16.	Ringolact solution	OTSUKA	234 K90	Not required	4	-	-
17.	Compound sodium Lactate solution	B. BRAUN	M033141	Not required	4	-	-
18.	Diasol 34 standard	BAXTER	GR131268	Not required	34	-	-
19.	HD concentrate	B. BRAUN	0361057/8	Not required	34	-	-
20.	0.9% NaCl 5% Dextrose	B. BRAUN	M945142	Not required	4	-	-
21.	Water for injection	B. BRAUN	M920111	Not required	-	-	-
22.	Water for injection	SQUIBB	1228	Not required	-	-	-
23.	Distilled water	PHARMA- CEUTICS LAB	311291	Not required	-	+	+
24.	Metronidazole Inj.	B. BRAUN	M935161	Required	13	-	-
25.	Abazole injection	ABBOTT	53-801-XV	Required	13	+	+

*Maximum valid dilution.

Table 3
Results of controls
Slide method

Positive control	I	II
0.25 EU/ml	+	+
0.25 EU/ml	+	+
0.125 EU/ml	-	+
0.0625 EU/ml	-	+
Negative control	-	-

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